Ask My Docs: pipeline measurements

This report records what the retrieval pipeline actually did when it was built, rather than what it was expected to do. Every figure here came from a recorded run on the committed sample dataset.

Embedding throughput

Embedding is the slowest stage of ingestion and the one most worth measuring, because it has to be repeated in full every time the chunk size changes. On this machine the local model embedded 155 chunks in 29.90 seconds, which is 192.9 milliseconds per chunk on CPU with no GPU acceleration.

That number is the reason the local embedding backend was chosen over a paid API. Re-embedding the whole corpus to compare one setting against another is free and offline, so there was never a reason to hesitate before running an experiment. Per-call billing makes a learner cautious, and caution is the one thing a project like this cannot afford.

Does the vector database return the same answers as a plain scan?

Before the vector database was introduced, search was a linear scan over a Python list: compare the question against every chunk, sort by similarity, take the top five. That version cannot miss a match, because it examines every candidate.

Chroma does not examine every candidate. Its HNSW index deliberately skips most comparisons, which is exactly where the speed comes from. That makes it an approximate search, not an exact one - a trade that is easy to get backwards, since a database sounds more rigorous than a for loop rather than less.

The two were compared on ten questions against the same corpus. They agreed on 10 of 10, returning the same top five chunks in the same order. At this corpus size the approximation costs nothing measurable. On a corpus large enough for the index to matter, it would.

Distance is not similarity

Chroma's default distance metric is squared L2, not cosine. The difference was measured directly with two perpendicular unit vectors: the default space reported a distance of 2.0 where cosine reported 1.0. For embeddings that are already normalised both metrics rank results in the same order, so configuring this wrongly raises no error and changes no answer - until a model that does not normalise its output is used, at which point retrieval quietly degrades.

Collections are therefore created with the cosine space set explicitly, and the retriever converts distance to similarity at the boundary so that every score the user sees means the same thing: higher is better.